

Run report

2026-10-04T034711Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	500,000
dim	4	dataset	gaussian
seed	46,522,706	data_seed	42
hidden_units	64	n_layers	2
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	3.077×10^{-3}	best_epoch	99
gaussian_params	dim=4, mu_gen=[1, 0, -0.5, 0.5], mu_true=[0.8, 0.1, -0.6, 0.7], cov_gen=[[1, 0.07, -0.22, 0.24], [0.07, 0.49, 0, 0.056], [-0.22, 0, 1.21, 0.616], [0.24, 0.056, 0.616, 0.64]], cov_true=[[0.64, 0, -0.24, 0.192], [0, 0.36, 0.12, 0], [-0.24, 0.12, 1, 0.3], [0.192, 0, 0.3, 0.36]], cov_detector=[[0.16, 0, 0, 0], [0, 0.25, 0, 0], [0, 0, 0.36, 0], [0, 0, 0, 0.09]]		
mmd_sigmas_detector	$1.590 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		
mmd_sigmas_particle	$1.336 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		

Timing

Phase	Seconds	Share (%)	Detail
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Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein Distance [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	231.7	13.71	94.1	28.77	87.6	28.84	87.5
dim_1	107.6	30.51	71.7	19.61	81.8	16.43	84.7
dim_2	116.2	15.51	86.7	44.05	62.1	70.69	39.2
dim_3	218.7	4.943	97.7	25.91	88.2	37.68	82.8

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	13.83	0.3508	97.5	0.5086	96.3	0.5821	95.8
dim_1	4.552	0.8811	80.6	0.4754	89.6	0.4076	91.0
dim_2	2.293	0.3275	85.7	0.5927	74.2	0.8021	65.0
dim_3	20.87	0.2593	98.8	0.4192	98.0	0.7307	96.5

VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	52.60	1.376	97.4	1.997	96.2	2.269	95.7
dim_1	17.80	3.449	80.6	1.868	89.5	1.595	91.0
dim_2	9.023	1.262	86.0	2.324	74.2	3.166	64.9
dim_3	78.36	1.019	98.7	1.658	97.9	2.891	96.3

Sliced Wasserstein, joint [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
<i>All</i>	227.5	121.2	46.7	30.79	86.5	42.53	81.3

Mean 1D distance over random directions, in Data standard deviations. IBU: product of its per-observable weights.

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	240.2	22.98	90.4	29.93	87.5	30.34	87.4
dim_1	117.8	48.63	58.7	26.57	77.4	20.30	82.8
dim_2	121.0	23.97	80.2	42.85	64.6	70.59	41.7
dim_3	226.5	17.58	92.2	21.69	90.4	35.96	84.1

JS divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	18.41	2.874	84.4	0.4936	97.3	0.6525	96.5
dim_1	9.436	4.348	53.9	0.9671	89.8	0.7966	91.6
dim_2	3.592	1.282	64.3	0.7791	78.3	1.095	69.5
dim_3	27.10	4.122	84.8	0.3375	98.8	0.8679	96.8

VLC divergence [$\times 10^3$]

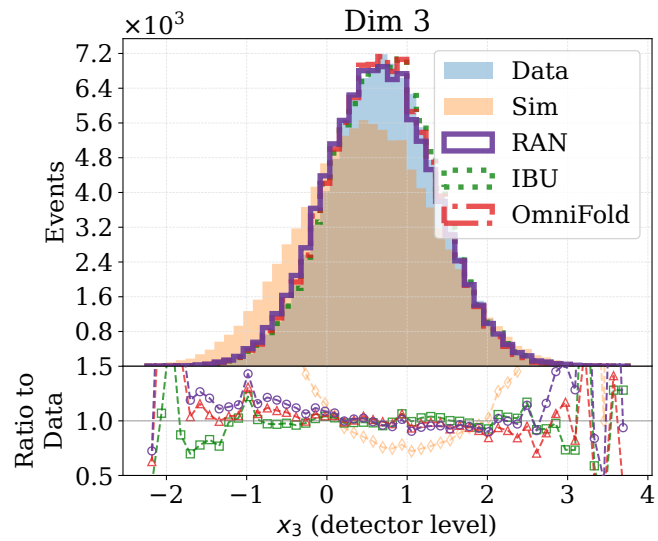
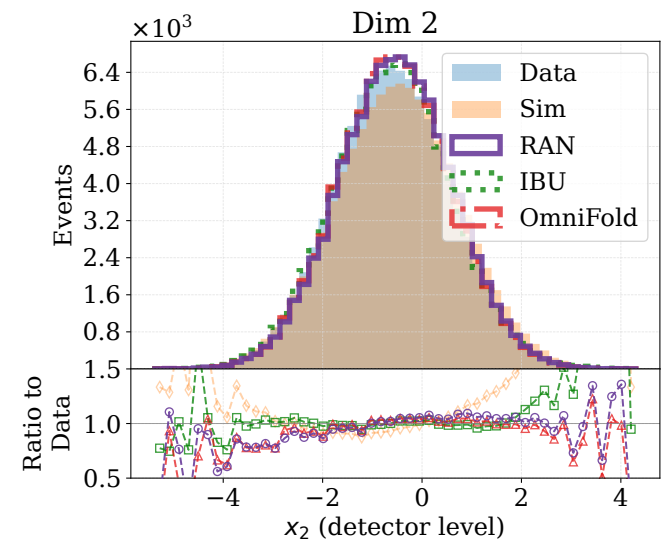
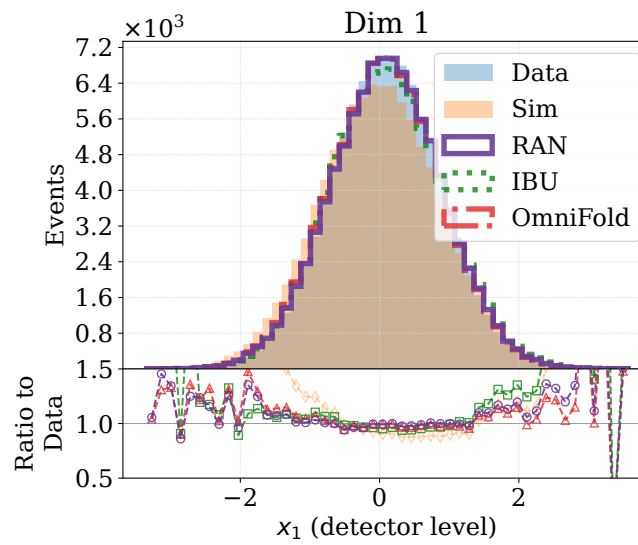
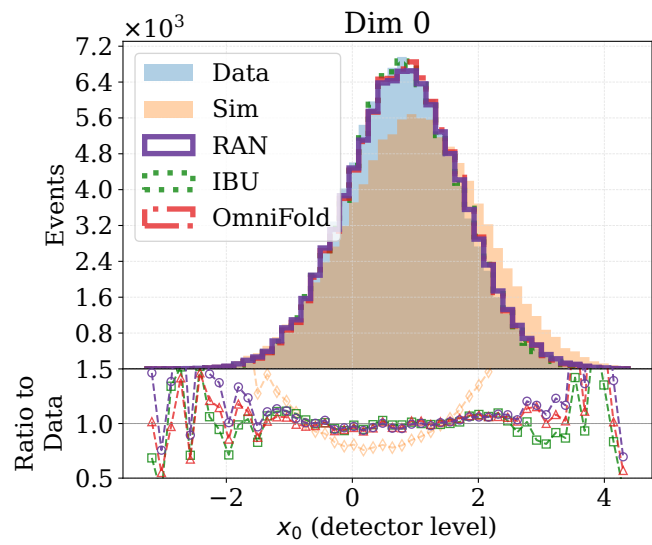
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
dim_0	69.53	11.39	83.6	1.947	97.2	2.559	96.3
dim_1	36.25	17.04	53.0	3.790	89.5	3.096	91.5
dim_2	14.07	5.021	64.3	3.057	78.3	4.335	69.2
dim_3	100.6	16.25	83.8	1.331	98.7	3.423	96.6

Sliced Wasserstein, joint [$\times 10^3$]

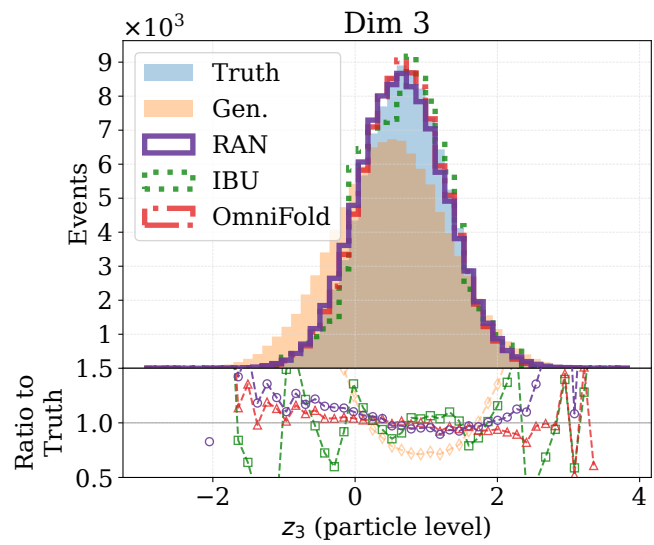
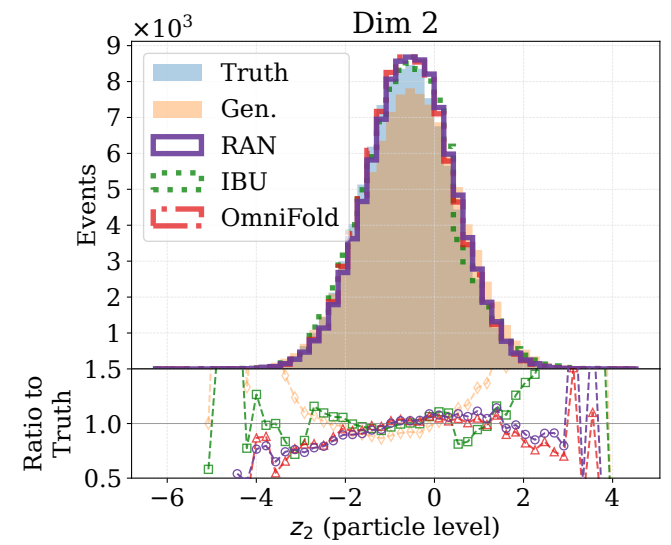
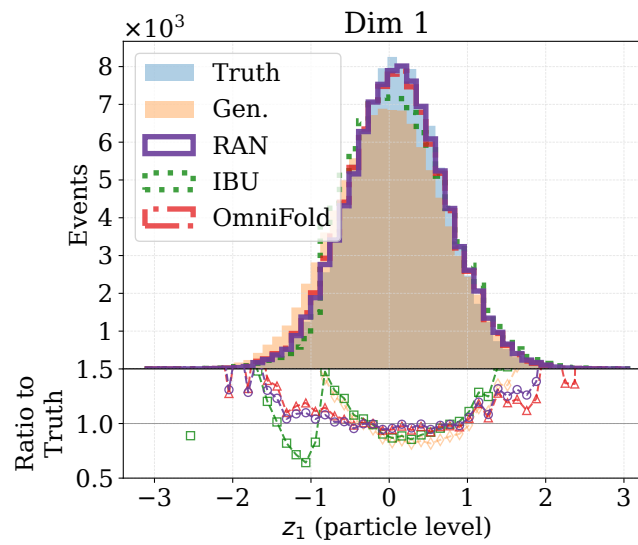
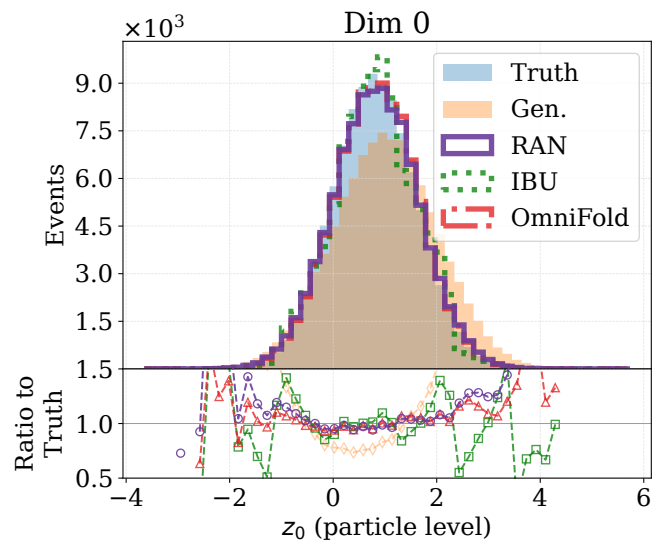
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
<i>All</i>	272.6	142.6	47.7	35.15	87.1	50.34	81.5

Mean 1D distance over random directions, in Truth standard deviations. IBU: product of its per-observable weights.

Detector Level

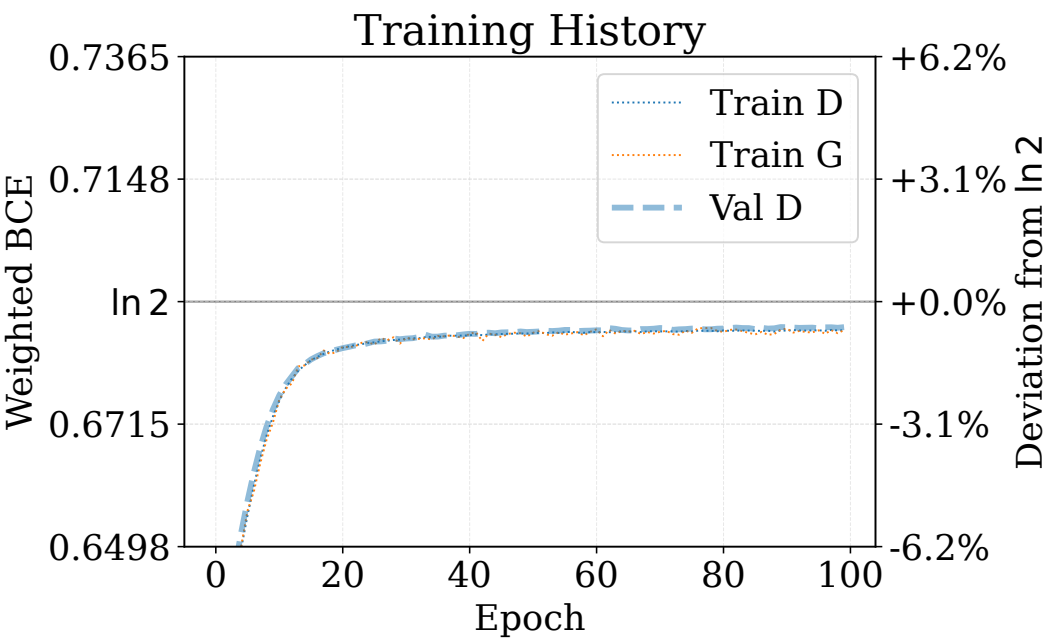


Particle Level



Training diagnostics

Training losses



Selection criteria

